

TPSA HACKATHON

TPSA²

Problem Sheet

Problem Context

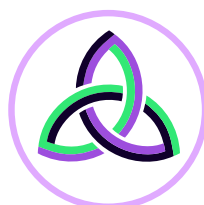
Difficulty Scalings
Data and Scripts

2

Ranked Problems

Recruit
Regular
Hardened
Veteran

3





Problem Context

Traffic modelling has long fascinated physicists and mathematicians. It's one of the most apparent uses for the complicated mathematical, physical, and statistical techniques associated with highly complex dynamical systems. Analogously to how nature generated complex real-world systems which seems eerily optimised, one might assume that there are more and less optimal ways to construct our own systems. The purpose of this hackathon is to investigate the potential optimisations processes applicable to the Irish transport network. Primarily the Dublin road system.

This investigation can be carried out through a number of lenses including but not limited to

- Linear algebra based simplifications
- Computational scaling
- Scalar field theory
- Statistical Physics
- Information Theory & Metrology

DIFFICULTY SCALINGS

What is expected of each difficulty level?	
Recruit	Very Likely Progress
Regular	Likely Progress
Hardened	Uncertain Progress
Veteran	Minor Progress is Impressive

DATA AND SCRIPTS

We have prepared some data and will be adding to it throughout the week. This can be found at the following github repo:

github.com/farrencc/Hackathons



Ranked Problems

RECRUIT

Clustering Capacity

Knowing the geometry of the network and speed of the roads, are there preferred veins of travel from a purely network capacity perspective?

- How might the problem be framed to make it suitable to a linear algebra-driven analysis?
- What would eigenvalues and eigenvectors represent in such a context?

Clustering Flow

With simulated data to hand, what are the preferred veins of travel in practice?

- How might travel volume data be aggregated in a manner that linear algebra can tackle?
- Do you notice any discrepancies between it and the routes stemming from the network geometry clustering?

Area/Zone Specific Model

In this hackathon, some of the later years will be attempting to build a model from scratch. Given that we only have 4 days, it's likely that a one-size-fits-all approach will be used. If we were trying to develop this further and we wanted the accuracy that a zone-specific model could bring (specific models for commercial vs. residential zones, or motorway dynamics), what models are out there, and what are their characteristics?

Pedestrianisation Analysis

Can you apply your spectral (eigenvalue- and eigenvector) analysis to investigating pedestrianisation opportunities?

REGULAR

Build The Model

Can you build a model from scratch to sensibly mimic traffic flow? It will need to include:

- Flow conservation at junctions.
- Source and sink terms to represent parking/leaving home/work/areas of interest.
- Potential-like attractive pulls during the working day / pushes during evening and night.
- Road speeds to inform max flow, and road lengths to inform max capacity



- (Interesting!) Density-based congestion effects.
- (Hard, but cool) Adaptive flow rates to respond to density.

Construct the Potential

At some point, we have to start informing the approach with real data!

- Can you reverse engineer what the potential should look like from the SATS data?
- How does parking availability play into this?

Discern The Boundary

Computational Scaling Analysis

If we had a fully operational traffic model, can you hazard a guess based on some experiments at how computational difficult it would be to scale?

- If you want to do a special zone analysis - say resolution down to bus, cyclist, car
- how much additional overhead does that bring?

HARDENED

Develop Correlation Functions

(Assumes model is built) It would be useful to know how roads that aren't immediate neighbours relate to one another in a traffic sense.

- Can you develop a correlation function that measures how two regions tend to flow/jam together?
- Can you implement this function programmatically?
- What is the protocol for ensuring the statistical significance of the found correlation function?

Jam Forecasting

Wouldn't it be cool to be able to pre-empt the jam, and adjust our traffic lights accordingly?

- Can a two-time correlation function approach be developed (and is it worth it/ is there a better framing)
- How would you affect the traffic lights to reflect this anticipated jam (and how could you be sure you're not making a worse one)?

Crisis Response

In the modern world we rely on the 24/7 availability of our critical infrastructure: electrical infrastructure, emergency services, road network, etc. to be resilient against unforeseen high stress events.



- Can you develop a means of testing the ability of Dublin's road networks to adequately handle and dissipate large volumes of traffic under differing scenarios?
- How much the lights be coordinated in the event of a major crisis?

VETERAN

Reverse Engineer the model

(Assumes model is built) In reality, we will have to try to pin a model to real data. Let's build up to that slowly:

- Simulate some data with the built model. How would you reverse engineer the model parameters if you only had the data?
- How does this change as you make the information imperfect (noise / imperfect sensor coverage of the network)? We understand this is fitting the output of a model back to the same model that produced it, and that it's somewhat contrived. However, in the absense of more granular data, this is our best shot!

Metrology

As will be known to any of you who have participated in a previous TPSA hackathon, it's the nature of tax-payer funded development that the optimal design for systems - including data collection - is often neglected in favour of more cost-effective routes. To combat this, we need to ensure that our experimental design i.e. our sensor placement, is as optimal as possible.

- Given perfect information and a readily built model, how might we find the optimal traffic sensor locations?
- Assume that our sensors yield locally perfect information as is effectively seen in Transport Infrastructure Ireland's Traffic Count Map.

Apply Metrology protocol elsewhere

The real test of the metrology protocol is to check how it runs in a different location.

- Take the traffic model, and apply it to Cork to simulate some data.
- Apply the Metrology protocol to determine where best to place a reasonable budget of sensors.
- Check to see if the model parameters can be successfully reverse engineered.